CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. (CL) Examination

November 2012

CL302 Structural Analysis-II

Date: 27.11.2012, Tuesday | **Time: 10:00 a.m. To 01:00 p.m.** | **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q – 1(a) The shear stress in a material at a point is given as 50 N/mm^2. Determine the local strain energy per unit volume stored in the material due to shear stress. Take C = 8 x 10^4 N/mm^2. **[07]**

Q - 2 (a) Calculate fixed end moments for a beam loaded as shown in **Figure 1** using theorem of minimum strain energy. **[05]**

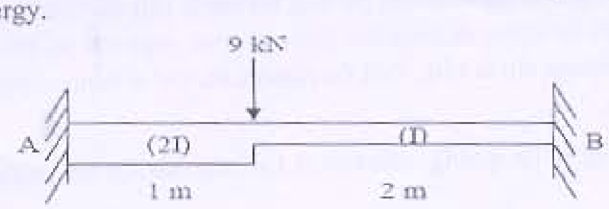

Figure 1

(b) Analyze **any one beam** from the beams shown in **Figure 2 & Figure 3** by consistent deformation method and draw S.F. and B.M. diagram. **[09]**

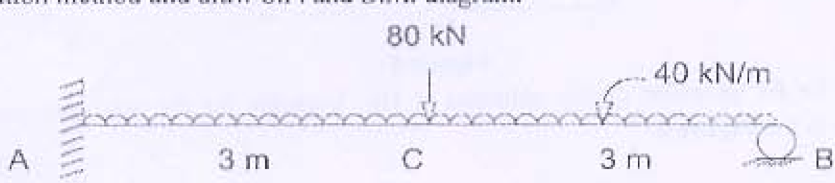

Figure 2

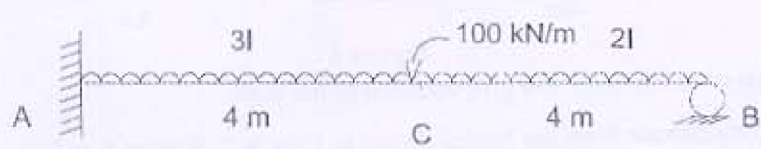

Figure 3

Q - 3 (a) Using Energy principle, calculate θ_B and δ_B for the beam shown in **Figure 4**. Take E = 2 x 10^5 N/mm^2, I = 5 x 10^8 mm^4. **[04]**

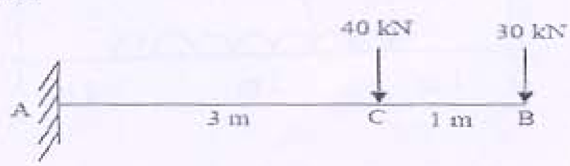

Figure 4

(b) **Attempt any Two.** [10]

(i) A thick metallic cylindrical shell of 150 mm internal diameter is required to withstand an internal pressure of 8 kN/mm^2. Find the necessary thickness of the shell, if the permissible tensile stress in the section is 20 N/mm^2.

(ii) A thick cylidrical pipe of internal radius of 12 cm and external radius 16 cm is subjected to an internal fluid pressure of 120 kg/cm^2. Determine the maximum and minimum hoop stresses in the cross-section. Also sketch the hoop stress distribution across the section. What is the percentage error if the maximum hoop stress is found from the equation for thin pipes.

(iii) A compound cylinder is composed of a tube of 250 mm internal diameter at 25 mm wall thickness. It is shrunk on to a tube of 200 mm internal diameter. The radial pressure at the junction is 8 N/mm^2. Find the variation of hoop stress across the wall of the compound cylinder, if it is under an internal fluid pressure of 60 N/mm^2.

SECTION – II

Q – 4(a) A cable supported at its ends 40m apart carries loads of 20 kN, 10 kN and 12 kN at distances 10 m, 20 m and 30 m from the left end. If the point on the cable where the 10 kN load is supported is 13 m below the level of the end supports. Determine (i) the reactions at the supports (ii) the tensions in the different parts of the cable (iii) the total length of the cable. [07]

Q - 5 (a) A cable loaded with 10 kN/m is stretched between two supports in the same horizontal line 200m apart. If the central dip is 15m, find the maximum and minimum pulls in the cables. [04]

(b) **Attempt any One** [10]

(i) Sketch ILD for M_X @ x=1m giving ordinates at 1m intervals for the propped cantilever shown in the **Figure 5**.

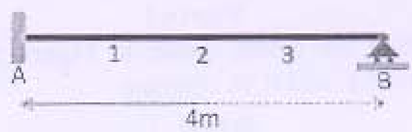

Figure 5

(ii) Sketch ILD for F_X @ x=2m giving ordinates at 1m intervals for the two span continuous beam shown in the **Figure 6**.

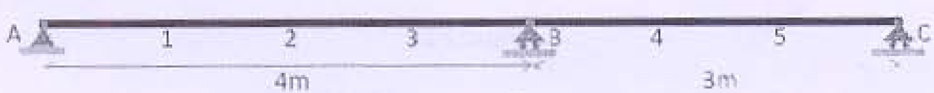

Figure 6

Q - 6 (a) State Muller-Breslau Principle and give equation of the same. [02]

(b) Analyze **any two beams** from the beams shown in **Figure 7, Figure 8** & **Figure 9** and draw S.F. and B.M. diagram. [12]

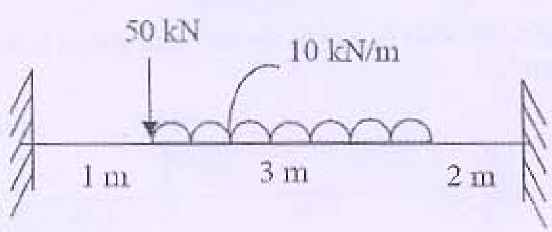

Figure 7

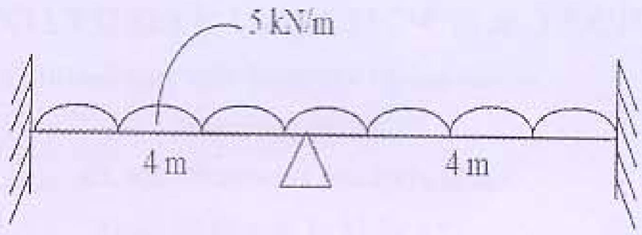

Figure 8

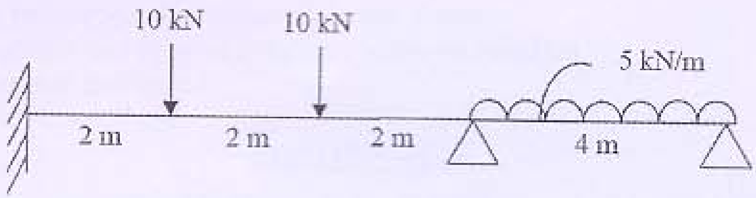

Figure 9

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